Mean, Median, and Baselines

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

Before you start

Week 4: mean divides a sum by the count; median is the middle sorted value.

Plan time to read, investigate and explain. Keep predictions separate from observations.

Your mission

> Start with the target: Fit mean and median constant baselines on training targets. Try the recall before opening notes.

Your learning target

I can fit **mean** and **median constant baseline**s on **training** targets.

I can compare constant **prediction**s on separate records.

I can keep fitted constants independent of evaluation answers.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

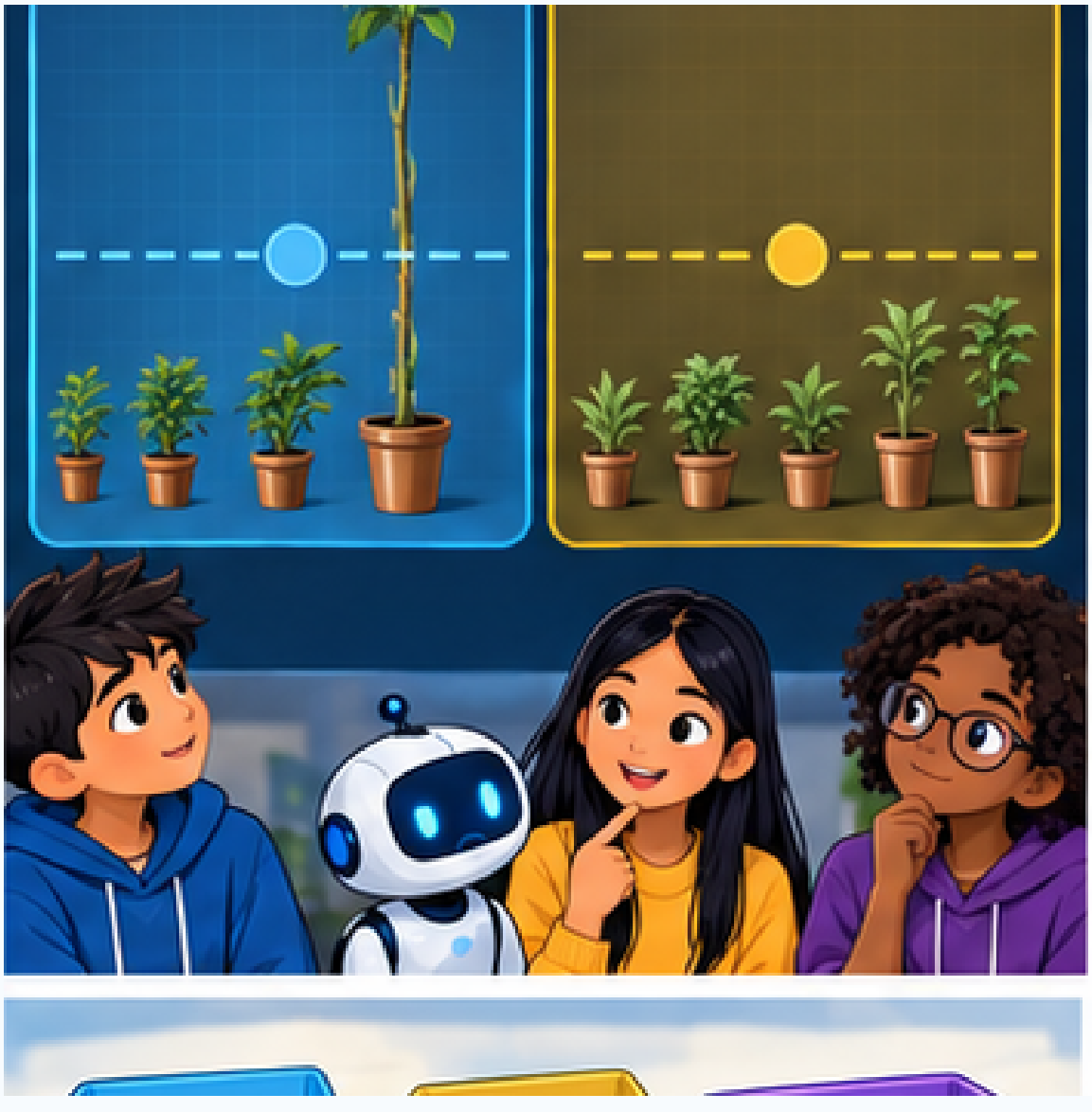
Without opening the old lesson, explain one key idea from Regression and Residuals. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Precision and Recall. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Scores and Calibration Intuition. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Mean and **median** are transparent regression baselines; **outlier**s affect them differently.



Your target

LEARNING TARGET For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

Builder name and experiment date:

Readiness for the connected explanation

Why can signed residuals hide error size?

This week's end goal

By the end, I can explain **mean**, **median**, and baselines using a traced example and a stated limitation.

STUDENT LESSON · 2 OF 18

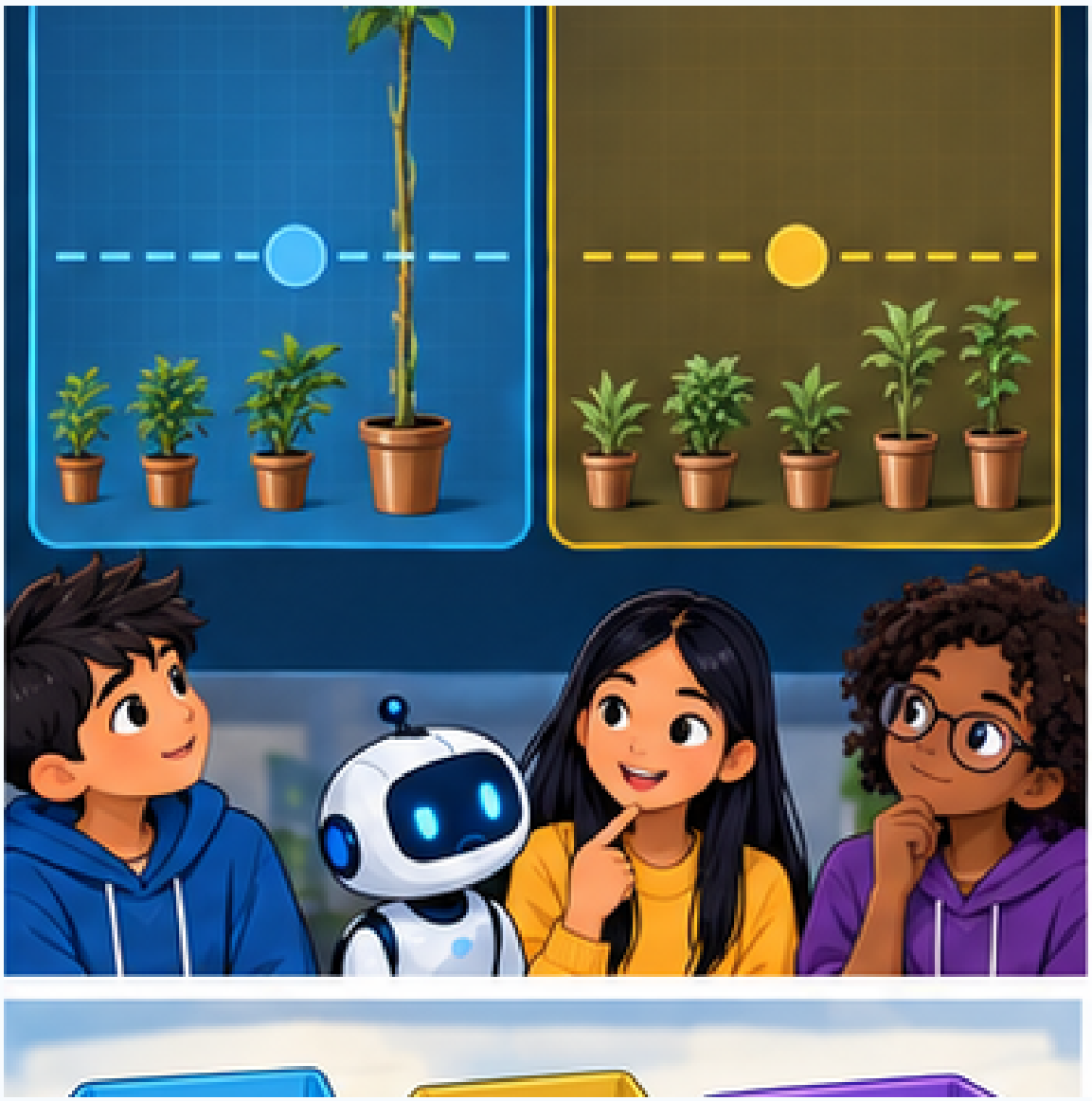
Notice the evidence

> Investigate the supplied Mean, Median, and Baselines case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible **evidence**; do not explain more than the picture and information support.



Start with this puzzle

MYSTERY The team must use **mean**, **median**, and baselines to solve a model-building problem. What information do they have? What important **evidence** is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what **evidence** would confirm or reject your **prediction**.

My **prediction** and supporting **evidence**:

A closer explanation

Before a complex regression model, fit a simple constant predictor. A **mean** baseline uses the **training** average. A **median** baseline uses the middle **training** target.

They ignore features but provide a comparison that is easy to reproduce. Evaluation targets must not determine the constants.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines constant **prediction** baselines.

STUDENT LESSON · 3 OF 18

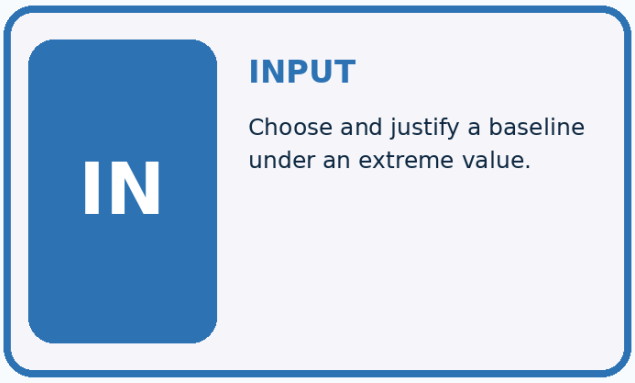
See how it works

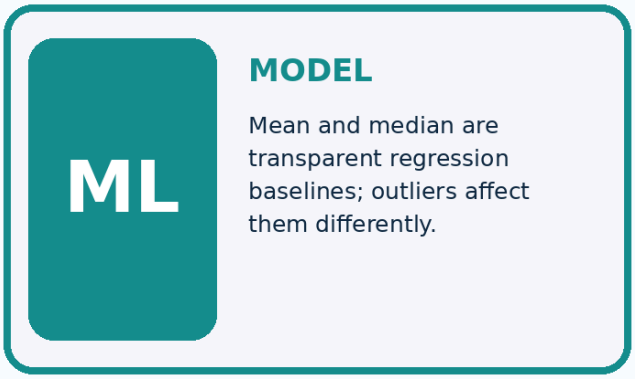
> Follow the mechanism for Mean, Median, and Baselines. Connect each labelled input, deciding step and result.

ML MECHANISM

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.







Question to explain

QUESTION Choose and justify a baseline under an extreme value.

MODEL IDEA **Mean** and **median** are transparent regression baselines; **outlier**s affect them differently.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

How the mechanism works

UNDER THE HOOD Heights are 10, 11, 12, 13, and 54 cm. **Mean** = 100 ÷ 5 = 20 cm.

Draw a fresh data → model → **prediction** → evaluation trace:

A closer explanation

Training targets [2, 4, 6, 12] have **mean** 24/4=6 and **median** (4+6)/2=5. On separate targets [4, 8], the **mean** baseline predicts [6, 6] with **absolute error**s [2, 2], averaging 2.

The **median** baseline predicts [5, 5] with errors [1, 3], also averaging 2.

```
from statistics import mean, median
training = [2, 4, 6, 12]
validation = [4, 8]
for name, value in [("mean", mean(training)),
                    ("median", median(training))]:
    mae = mean(abs(y - value) for y in validation)
    print(name, value, mae)
# Both validation MAEs are 2; constants differ.
```

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

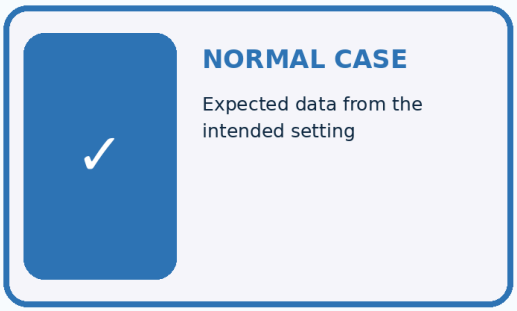
Compare the cases

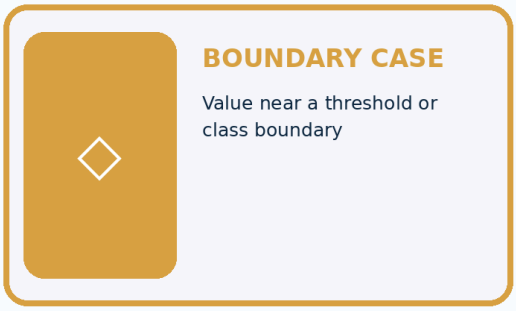
> Compare the supplied cases for Mean, Median, and Baselines. Identify the detail that changes the conclusion.

EVALUATION DESIGN

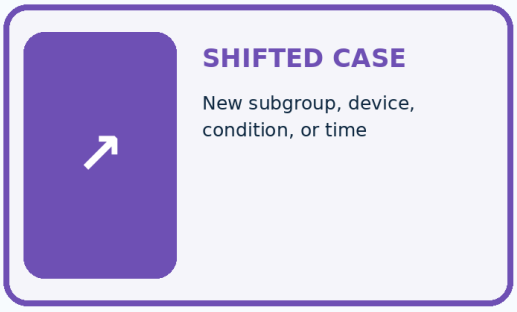
Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.









NORMAL Expected data and expected behavior.

Keep this boundary clear

BOUNDARY Value close to a threshold, ambiguous label, or uncertain **prediction**.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change one thing

Change: Replace **training** 20 with 8 while keeping evaluation unchanged.

Predicted result

Prediction: Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

Why the result follows

Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

> Use the vocabulary of Mean, Median, and Baselines to explain a concrete decision, rather than only memorising definitions.

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.

1

MEAN

sum divided by count

2

MEDIAN

middle ordered value

3

OUTLIER

value far from most observations

4

LOCATION BASELINE

single typical value predicted for all cases

Term	Precise meaning	My ML example
mean	sum divided by count	
median	middle ordered value	
outlier	value far from most observations	
location baseline	single typical value predicted for all cases	

MISCONCEPTION TO REPAIR The **mean** is always the fairest and most scientific summary.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word	Meaning
Constant baseline	One fixed prediction for every query
Training mean	Sum of training targets divided by their count
Training median	Middle value of sorted training targets
Absolute error	Magnitude of observed minus predicted

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Mean, Median, and Baselines, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

WEEK 20 • WORKED EVIDENCE

1. Heights are 10, 11, 12, 13, and 54 cm.
2. Mean = 100 ÷ 5 = 20 cm.
3. Median is 12 cm, the middle ordered value.
4. For typical-height absolute error, median is more robust; still report why 54 is present.

Step 1

STEP 1 Heights are 10, 11, 12, 13, and 54 cm.

Step 2

STEP 2 **Mean** = 100 ÷ 5 = 20 cm.

Step 3

STEP 3 **Median** is 12 cm, the middle ordered value.

Step 4

STEP 4 For typical-height **absolute error**, **median** is more robust; still report why 54 is present.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the **evidence** independently:

A closer explanation

Constant prediction baselines at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Training targets	Fit mean 6	Freeze constant
Training targets	Fit median 5	Freeze constant
Evaluation targets	Compare fixed predictions	Both MAE 2 here

Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

Calculate MAE for both baselines on [4, 8]. Explain the steps or evidence behind your response.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Mean, Median, and Baselines

Fit two constant height baselines and compare them on held-out examples.

The facts we will use

Supplied facts

Training heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/20

Mean, Median, and Baselines

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Fit mean

Training mean=(2+4+6+20)/4=8 cm.



2. Fit median

Training median=(4+6)/2=5 cm.



3. Evaluate constants

Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm.

CHANGE ONE THING

Replace training 20 with 8 while keeping evaluation unchanged.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Fit mean	Training mean=(2+4+6+20)/4=8 cm.
2. Fit median	Training median=(4+6)/2=5 cm.
3. Evaluate constants	Mean MAE=(4+2)/2=3 cm; median MAE=(1+5)/2=3 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

Keep the stated limit

Mean and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

Keep the stated limit

Read the labelled picture: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Keep the stated limit

Change one thing: Replace **training** 20 with 8 while keeping evaluation unchanged. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Mean, Median, and Baselines claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.

!

COMPARE • DIAGNOSE • REPAIR

Tempting claim: The mean is always the fairest and most scientific summary. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record	Expected	Actual	Likely cause / next test
normal case	passes		
boundary case	defined behavior		
missing case	safe fallback		
shifted case	measured limit		

CLAIM TO CHALLENGE The **mean** is always the fairest and most scientific summary.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

For an even number of values, this course averages the two middle values to report one **median**. Keep a frozen baseline across comparisons and show its evaluation score.

A model that fails to beat it may still reveal a data or metric issue worth investigating.

Why not predict the evaluation mean as the baseline?

STUDENT LESSON · 8 OF 18

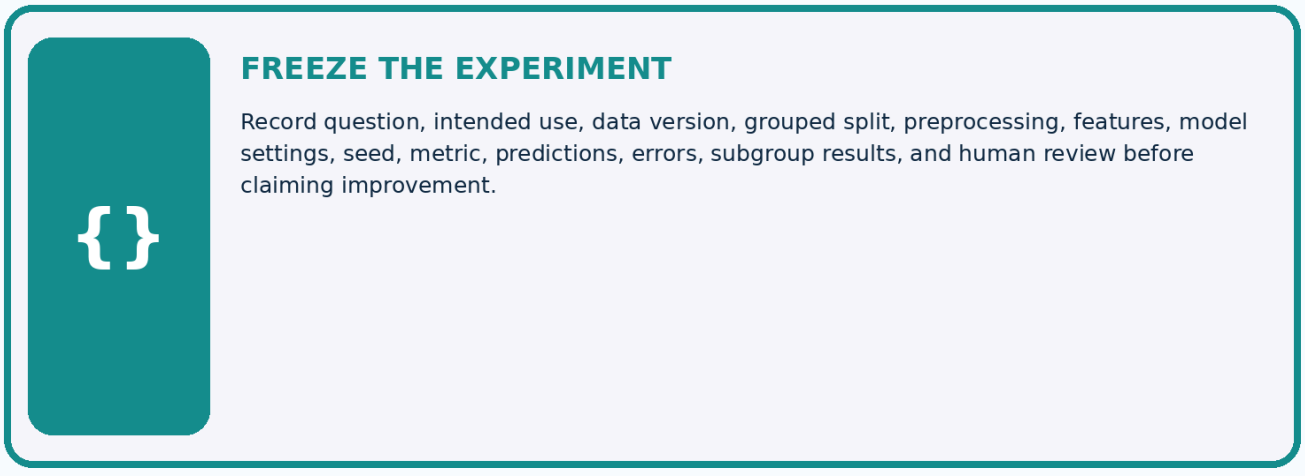
Practise together

Practise Mean, Median, and Baselines with support, then explain the deciding step in your own words.

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



Experiment record

> **Question to explain**
>
> QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/pre**process**ing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

Replacing **training** 12 by 40 gives **mean** 13 and **median** 5. That change is a data change to record and investigate, not an instruction to delete an inconvenient **observation**.

Decide whether the extreme value is valid using the measurement policy and provenance.

Practise with less help: Score separately

Use workbook W2 for the connected example “Constant **prediction** baselines”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Constant **prediction** baselines. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Mean, Median, and Baselines. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Baseline Tournament: compute mean and median, predict every case, compare MAE, and explain the outlier's effect.

1. Write the hypothesis and expected **evidence** before running.

2. Baseline Tournament: compute **mean** and **median**, predict every case, compare MAE, and explain the **outlier**'s effect.

3. Run one normal, one boundary, one missing, and one shifted case.

4. Calculate the metric from raw counts or row-level errors.

5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

> **Case:** Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

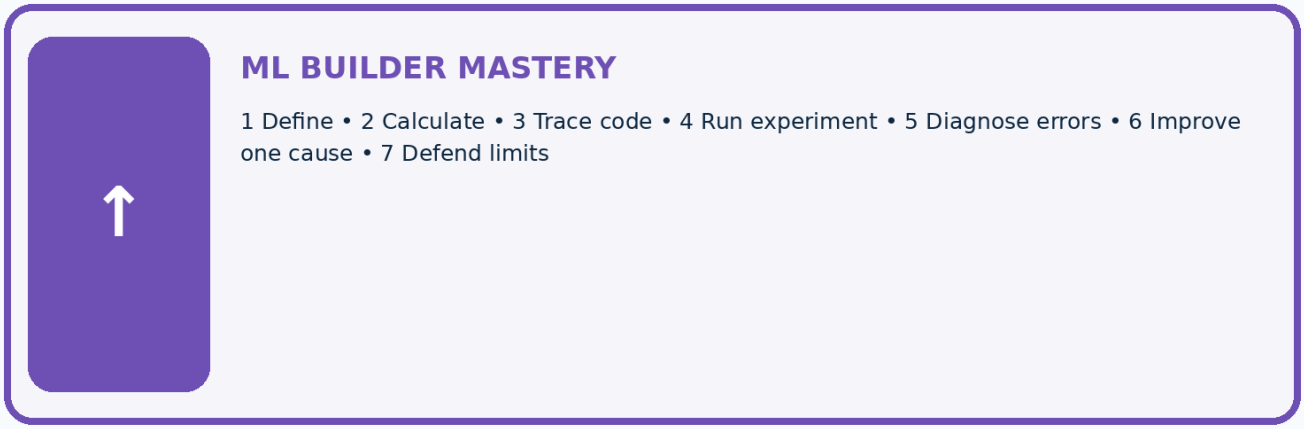
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Mean, Median, and Baselines. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the example.



DEFINE Use all four terms precisely: **mean**, **median**, **outlier**, **location baseline**.

CALCULATE Reproduce the worked **evidence** with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE The **mean** is always the fairest and most scientific summary.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed **evidence**.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Responsible Plant Classifier. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same **evidence**-based capstone.

RESPONSIBLE PLANT CLASSIFIER

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and outlier decision.

THIS WEEK'S CHECKPOINT Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My **evidence**-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

STUDENT LESSON · 12 OF 18

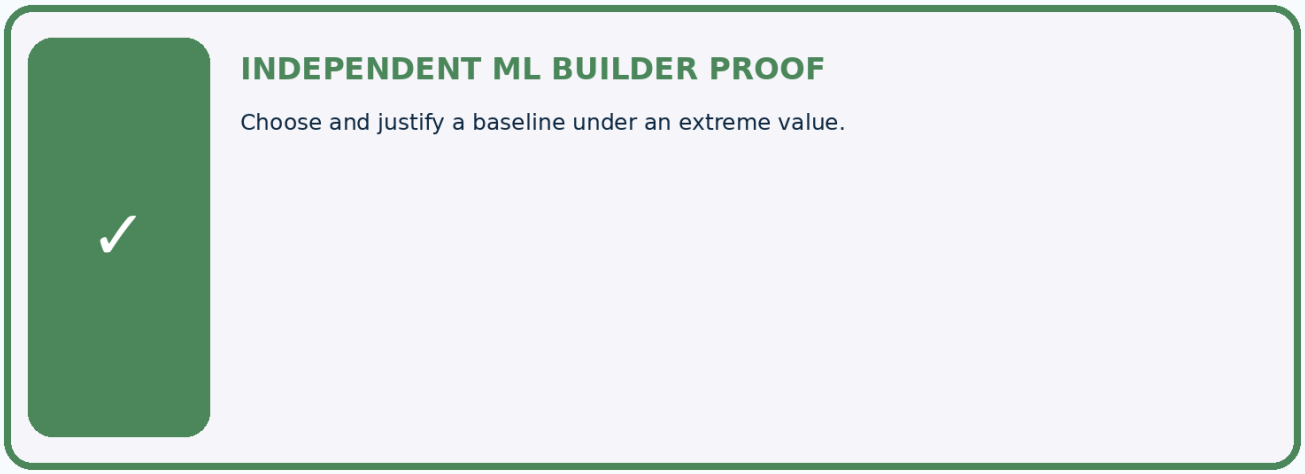
Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



What success looks like

REQUIRED PROOF Choose and justify a baseline under an extreme value.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent **evidence** and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

Connect to the next lesson

NEXT BRIDGE Carry the experiment **evidence** into: **Absolute Error** and MAE.

Study a complete case

> Revisit the complete Mean, Median, and Baselines case for supported practice. Repeated examples do not become fresh assessments.

Week 20 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Training heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

The question

Compute both and **training absolute error**s.

Worked explanation

Mean=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

Explain it in your own words

> Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Mean and median baselines use the same observations

Evidence or stage	Value or result
Observation A	2
Observation B	4
Observation C	12
Mean baseline	6
Median baseline	4

Training heights are 2,4,12 cm. Compare constant **mean** and **median** baselines.

Compute both and **training absolute error**s.

Mean=6, **median**=4. **Mean** errors 4,2,6 total12; **median** errors 2,0,8 total10. **Median** minimises **absolute error** here.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply independently

Training [1, 3, 5, 15], evaluation [3, 9]. Fit both constants and calculate MAE.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 20 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

For **training** heights 2, 3 and 10 cm, calculate **mean** and **median**. Which constant is less influenced by the high value?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Document any numeric baseline used in Plant Classifier feature checks, including unit, distribution, and **outlier** decision.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

For **training** targets [1,3,11], compute **mean** and **median constant baseline**s. Compare total absolute **training** error for each. Where must final performance be measured?

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Mean, Median, and Baselines, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Fit two constant height baselines and compare them on held-out examples.

By the end, I can	How I will show it
Fit mean and median constant baselines on training targets	Explain the deciding step, show a trace or test, and state one limit.
Compare constant predictions on separate records	Explain the deciding step, show a trace or test, and state one limit.
Keep fitted constants independent of evaluation answers	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Mean, Median, and Baselines without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Fit **mean** and **median constant baseline**s on **training** targets

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Fit two constant height baselines and compare them on held-out examples.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

1. Fit **mean**

2. Fit **median**

3. Evaluate constants

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Evaluate constants **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace **training** 20 with 8 while keeping evaluation unchanged. Which conclusion is supported by the supplied facts?

1. Both fitted constants become5 cm and both evaluation MAEs remain3 cm.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

Repair a missing building block

Review Regression and Residuals (ai-4/19). Goal: Distinguish numeric **prediction** from classification. Short practice: Which statistic reveals the size of errors without cancellation?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Precision and Recall (ai-4/17). Goal: Choose the denominator that answers the question. Short practice: Were the original precision and recall denominators the same?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Scores and Calibration Intuition (ai-4/14). Goal: Interpret calibration using groups of forecasts. Short practice: Does a calibrated 0.8 score **mean** every such leaf is spotted?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

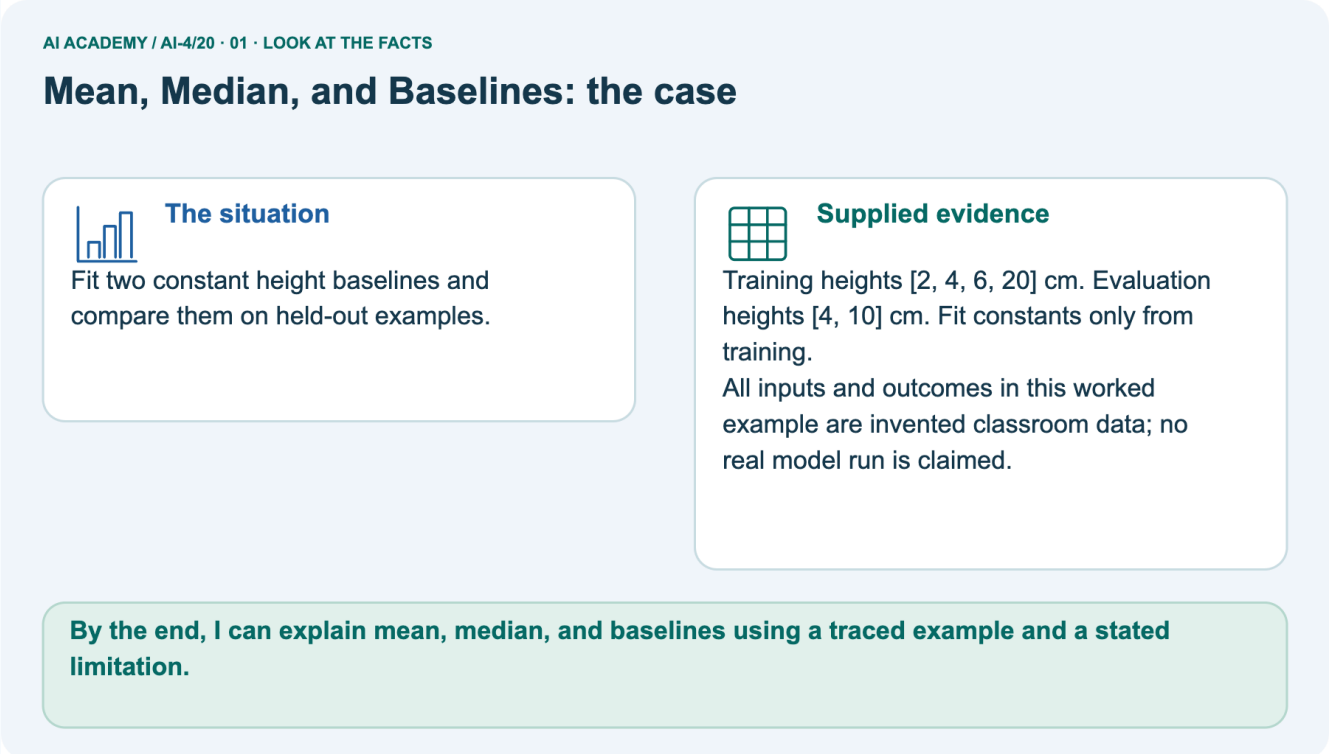
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Mean, Median, and Baselines graphic using its labels, deciding evidence and relevant limitation.

01. Mean, Median, and Baselines: the case



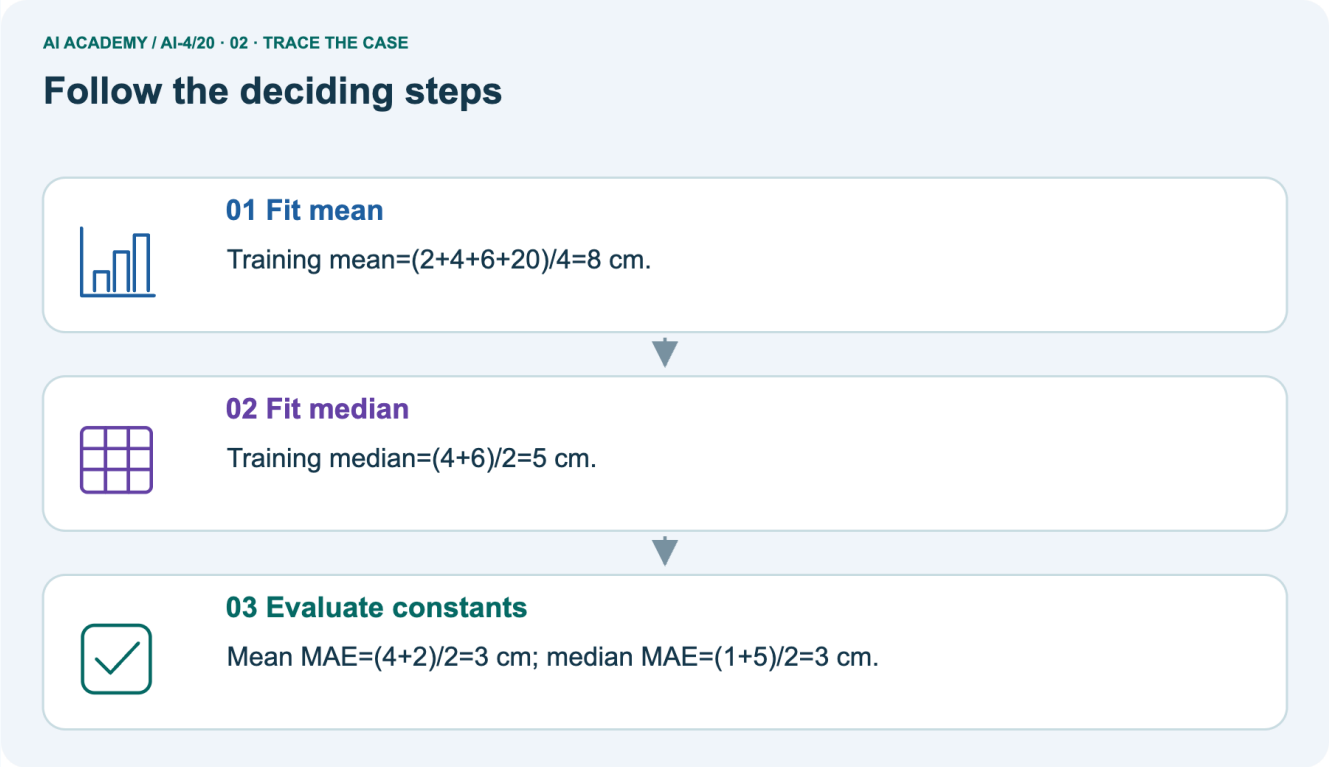
Follow the labelled picture with the matching explanation. Open the vector diagram

Fit two constant height baselines and compare them on held-out examples. Supplied facts: **Training** heights [2, 4, 6, 20] cm. Evaluation heights [4, 10] cm. Fit constants only from **training**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Fit **mean**: **Training mean**=(2+4+6+20)/4=8 cm. → Fit **median**: **Training median**=(4+6)/2=5 cm. → Evaluate constants: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm.

Point to each numbered stage and explain which recorded **input** it uses.

03. Constant prediction baselines at a glance

AI ACADEMY / AI-4/20 · 03 · INSPECT THE LESSON MODEL

Constant prediction baselines at a glance

Part	Evidence or action	Interpretation
Training targets	Fit mean 6	Freeze constant
Training targets	Fit median 5	Freeze constant
Evaluation targets	Compare fixed predictions	Both MAE 2 here

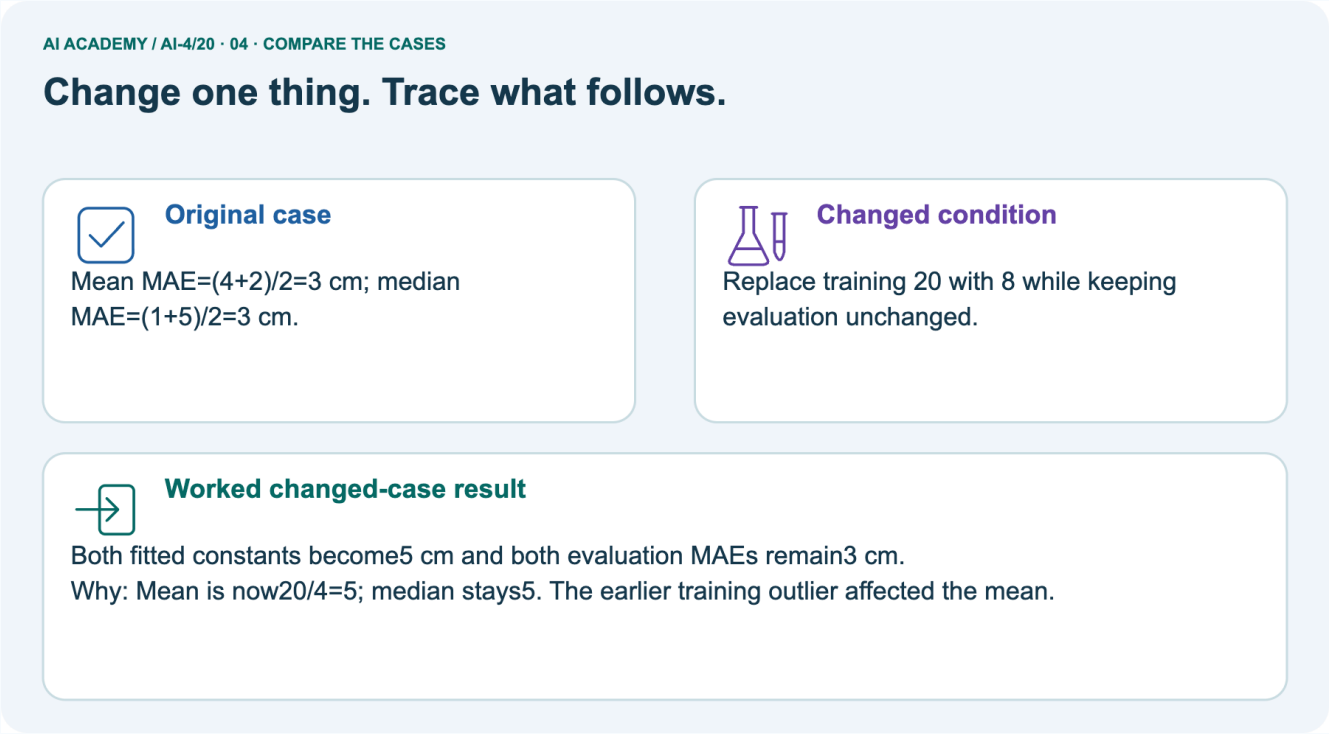
Mean and median answer different training objectives. A mean minimizes total squared error among constants; a median minimizes total absolute error. This does not guarantee either wins on a new evaluation set. With extreme values, the mean can move much more than the median.

Follow the labelled picture with the matching explanation. Open the vector diagram

Constant **prediction** baselines at a glance. **Training** targets | Fit **mean** 6 | Freeze constant. **Training** targets | Fit **median** 5 | Freeze constant. Evaluation targets | Compare fixed **prediction**s | Both MAE 2 here

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



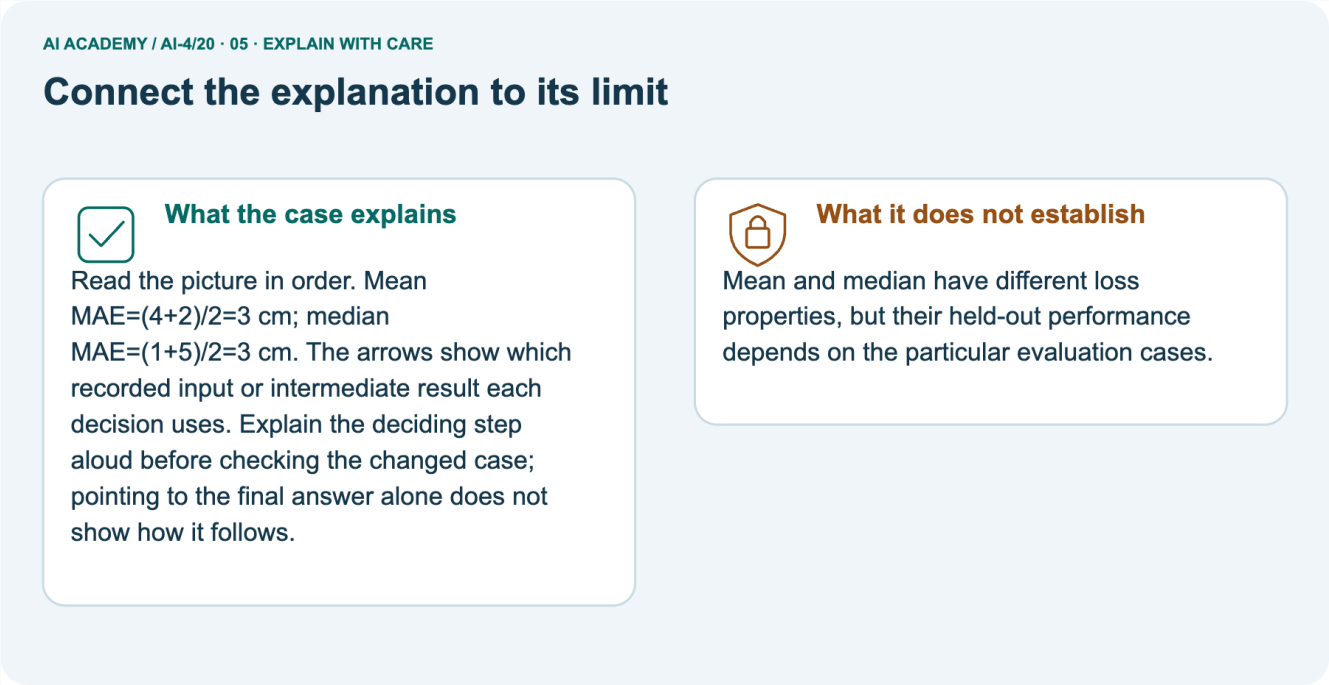
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. Changed condition: Replace **training** 20 with 8 while keeping evaluation unchanged.

Expected result: Both fitted constants become5 cm and both evaluation MAEs remain3 cm. Reason: **Mean** is now20/4=5; **median** stays5. The earlier **training outlier** affected the **mean**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

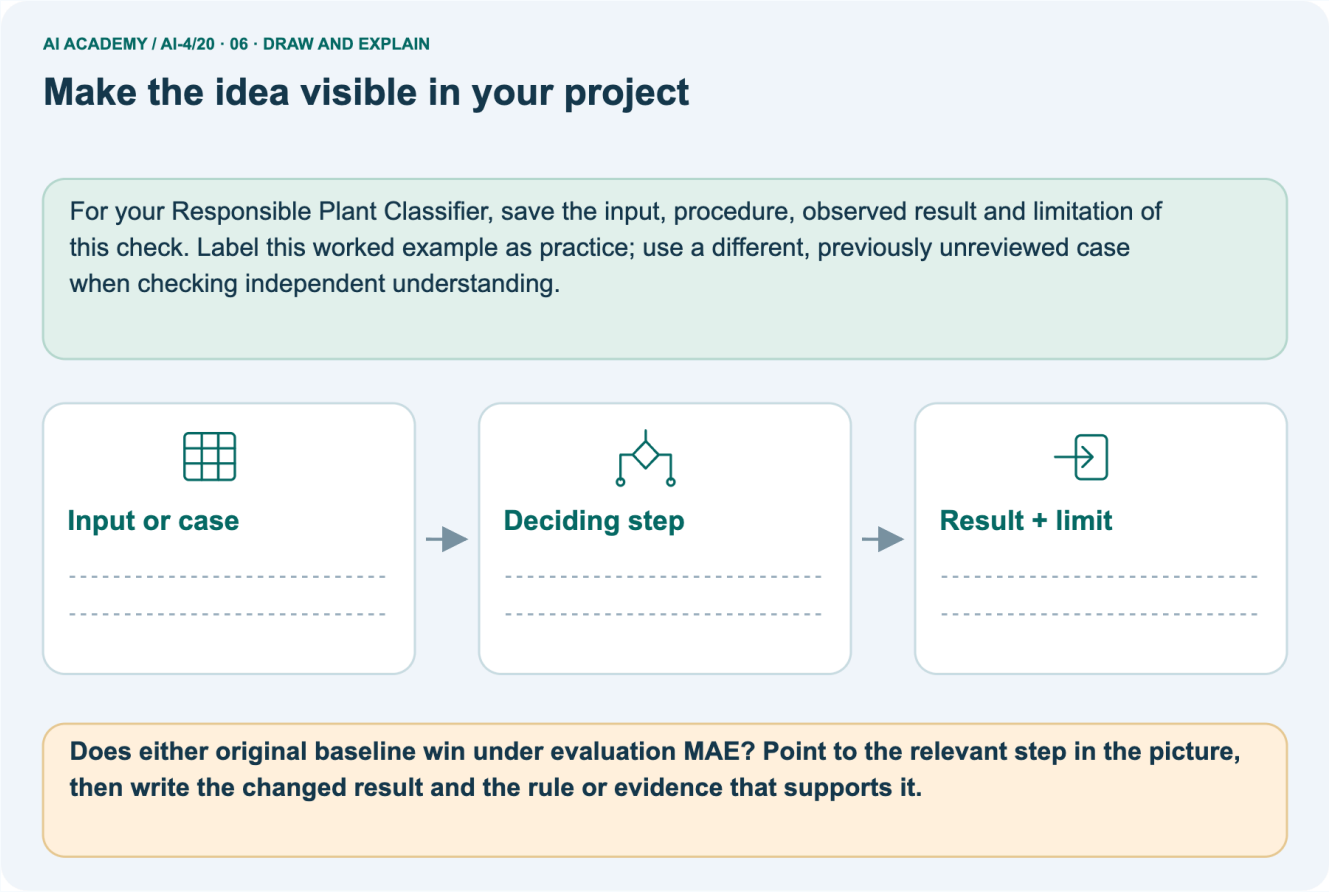
Read the picture in order. **Mean** MAE=(4+2)/2=3 cm; **median** MAE=(1+5)/2=3 cm. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: **Mean** and **median** have different loss properties, but their held-out performance depends on the particular evaluation cases.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does either original baseline win under evaluation MAE? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.